



Atmospheric pollution and environmental noise control in the city of Albacete

CLIENT

Albacete City Council

SECTOR

Urban

LOCATION

Albacete, Spain

PRODUCTS

Kunak AIR Pro
Kunak AIR Cloud
Kunak Noise

Need

The city council of Albacete, within its Sustainable and Integrated Urban Development Strategy (EDUSI), needed to deploy an air quality monitoring system using sensors to monitor the levels of air pollution and environmental noise control in the city and thus achieve one of its strategic objectives of reducing pollution levels in the city.

The monitoring network registers the main parameters to alert when any of them are altered through a notification system and thus help decision-making. In the same way, it will contribute to the fulfilment of the Sustainable Development Goals (SDGs) of the United Nations.

In addition, it provides greater transparency for the population as it includes visible information panels that will inform citizens in real-time of the different

indicators measured in a simple and intuitive way, as well as in the Environment section of the municipal website.

The project involves an investment of €271,764, 80% of which is financed with European funds through the Sustainable and Integrated Urban Development Strategy (EDUSI) supported by ERDF funds.





Albacete City Council [...] needed to implement a sensor monitoring network to monitor air pollution levels and environmental noise control in the city.



Solution

Deployment of a monitoring network of 10 Kunak AIR Pro stations that include visible information panels that will inform citizens in real-time in a simple and intuitive way of the different indicators measured and distributed throughout those areas of the city with the highest traffic density, thus providing information on the AQI (Air Quality Index).

After a preliminary study to determine the optimal locations, the air quality monitoring stations, which complement the information provided by the only measuring reference station in place until now, were located in the city's main arteries and in the vicinity of hospitals.



Results

This information provided by the sensor network will be of great help in detecting and taking measures against possible pollution scenarios that occur in the city when any parameter in the limit values or the quality objectives that have been established is exceeded.

With the information collected by these air quality stations, the elaboration of an emissions map is being considered, which will determine where to place, for example, new green areas of trees, the best locations to create Low Emission Zones (LEZ) or new green areas as carbon sinks.

This action contributes to achieving the environmental and cultural heritage conservation objective of Albacete's Sustainable and Integrated Urban Development Strategy (EDUSI), which is financed with resources from the European Regional Development Fund (ERDF). Thanks to the EDUSI funds, a municipal Smart City platform is also being set up for the smart management of the city, which will be fed, among other sources, by the data provided by these sensors.

Benefits

These data will be used, inter alia, for:

- Elaboration of zonal maps by pollution indicators.
- Identification of critical points.
- Carrying out vehicle traffic management campaigns.
- Adoption of corrective measures in the field of urban mobility.